SCALE-COVARIANT ATTENTION FOR TRAINING-FREE HIGH-RESOLUTION DIFFUSION TRANSFORMERS

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ABSTRACT

Diffusion transformers sampled far beyond their training resolution need attention that is cheap and stays inside the training distribution. Recent training-free methods give every head the same hand-scheduled receptive field. We ask FLUX.1-dev, at its training resolution, how much of each head's output a native-size window explains and how much a downsampled view of the whole image explains. A per-head, per-noise-level convex mix of these two in-distribution branches fits full attention closely. Its weight is stable across prompts (split-half correlation 0.99) and transfers across emulated scales and calibration resolutions at almost no cost. Scale-Covariant Attention (SCA) reuses this table at the target resolution. For the first 8 of 28 steps, while only layout-scale structure carries signal, every head uses the global branch alone, whose keys are pooled onto the native grid with rescaled coordinates. Afterwards each head mixes the global branch with an exact native-extent sliding-window attention by its calibrated weight. SCA needs one calibration pass and no training, and its attention cost is linear in the token count. On 191 Aesthetic-4K prompts at 4096^2, under one protocol shared with AReA's released code, SCA raises ImageReward from 0.92 to 1.23 and HPSv2.1 from 0.272 to 0.297, lowers FID from 146.0 to 138.4, and is significantly better on every preference, alignment, and no-reference quality metric we report. At 71 s per image on one GH200 it is 2.6 times faster than full attention and 3.0 times faster than AReA. The advantage grows with the scale factor: SCA ties AReA at 2048×4096 and raises ImageReward from 0.39 to 0.66 at 2048×6144. Both parts of the method are needed: without the opening phase subjects duplicate, and replacing the per-head mix with local-only attention after it lowers ImageReward by 0.36, because the mix is what carries global context through the remaining steps.

1 INTRODUCTION

Text-to-image diffusion transformers such as FLUX.1 (Black Forest Labs, 2024) and Stable Diffusion 3 (Esser et al., 2024) are trained at about one megapixel. Sampling them at 4096×4096 multiplies the number of image tokens by 16 and the cost of full attention by about 256. Speed is the smaller problem. Every attention layer then sees a token count and a range of relative positions outside its training distribution, and the outputs show duplicated subjects, broken layouts, and blurred texture (He et al., 2023; Du et al., 2023; Issachar et al., 2025; Rajabi et al., 2026a).

Most training-free fixes edit either the positional encoding (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2025; Rajabi et al., 2026b) or the set of keys each query sees (Zhang et al., 2023; Ma et al., 2025; Xue et al., 2026; Rajabi et al., 2026a). AReA (Rajabi et al., 2026a) is the most recent of the second kind. For the first 8 of 28 steps, full-resolution queries attend to keys and values subsampled onto a native-size grid, with query coordinates divided by the pooling factor; afterwards, queries attend inside windows that shrink geometrically to the native size, with a wavelet split test, a halo ring, random shifts, and a global key-value bridge to repair windowing artifacts. Every head of every layer receives the same receptive field at a given step, and the schedule is set by hand.

We question the shared receptive field. AReA's evidence for it, the narrowing of attention during denoising, is an average over heads and layers. We ask the pretrained model instead: at its training resolution, where full attention is trustworthy, how much of each head's output does a native-size

neighborhood explain, and how much does a downsampled view of the whole image explain? A convex mix of the two explains each head well (Section 4). The weight differs strongly between heads, barely changes across prompts, and changes little with the emulated scale or the calibration resolution. We fit it once per model in closed form against the model's own full attention.

Output matching cannot set everything. Applied from the first step at 4096^2, the calibrated mix duplicates subjects: one prompt for a Moai statue gives five statues (Figure 2). At high noise only the lowest spatial frequencies carry signal, and these set layout and object count. A native window covers a quarter of the image width at 4096^2, cannot resolve those frequencies, and invents a layout per window. Squared output error does not register this, because texture dominates it. SCA therefore runs the global branch alone while $\sigma > 0.89$ (the first 8 of 28 steps) and the calibrated mix afterwards (Figure 1). The opening phase coincides with AReA's; the two methods differ in the remaining 20 steps.

The ablations separate the two parts. The opening phase is essential: without it, subjects repeat. After it, the calibration says that most heads lean local (mean weight 0.64 to 0.74), yet forcing every head to local-only attention lowers ImageReward by 0.36 and HPSv2.1 by 2.9 points. The minority weight each head puts on the global branch is what keeps global context in the last 20 steps. An earlier configuration with a low-band guard hid this, because the guard injects the global branch's low band into every head whatever its weight.

Our contributions are:

- A protocol that emulates resolution extrapolation inside the training distribution and scores any receptive-field rule by its output error against full attention, from statistics collected in one pass.
- Evidence on FLUX.1-dev that receptive-field needs are head-specific, prompt-stable, and close to invariant under changes of scale and of calibration resolution.
- SCA, a training-free attention processor: a global-only opening phase, then a calibrated per-head mix of a native-grid global branch and an exact FlexAttention sliding window. One attention layer at 4096^2 takes 18.7 ms against 87.3 ms for full attention.
- An evaluation on Aesthetic-4K at 4096^2, 2048×4096, and 2048×6144 under one protocol shared with AReA's released code, with ablations of every component. SCA beats AReA at 4096^2 and 2048×6144, matches it at 2048×4096, and is 2.7 to 3.0 times faster than AReA. AReA's paper reports ImageReward 1.32 at 4K under its own settings, so numbers across papers are not comparable.

2 Related work

Training-free high-resolution generation. For U-Nets, ScaleCrafter dilates convolutions (He et al., 2023), HiDiffusion restructures the network and uses shifted windows (Zhang et al., 2023), FouriScale constrains the spectrum (Huang et al., 2024), and Jin et al. (2023) rescale the attention temperature. DemoFusion (Du et al., 2023), FAM (Yang et al., 2024), and FreeScale (Qiu et al., 2024) fuse a global path with local patches; I-Max (Du et al., 2024), HiFlow (Bu et al., 2025), and ScaleDiff (Koh et al., 2025) guide sampling with a native-resolution pass. Another line edits rotary embeddings (Su et al., 2021): Lumina-Next (Zhuo et al., 2024), DyPE (Issachar et al., 2025), UltraImage (Zhao et al., 2025), SigMa (Zhao et al., 2026), SEGA (Rajabi et al., 2026b), and TIDE (Liu et al., 2026). The closest relatives of SCA change which keys a query sees. ResDiT fuses a global branch with scaled positions and patch-wise attention by Gaussian and Fourier weights (Ma et al., 2025). HRDiT gives each head a static local scope chosen by an integer program under a cost budget (Xue et al., 2026). AReA schedules a shared receptive field from pooled-global to shrinking-local (Rajabi et al., 2026a). SCA differs from all three at once: its weights are per head and per noise level, its global branch is a zoom (pooling plus coordinate rescaling), and its weights are read off the model's own full attention instead of being set by hand or searched against a cost proxy.

Efficient and head-specific attention. ToDo attends from full-resolution queries to downsampled keys (Smith et al., 2024), the topology of our image-anchored branch without coordinate rescaling. DiTFastAttn selects window attention per layer and step (Yuan et al., 2024); for video, Sparse

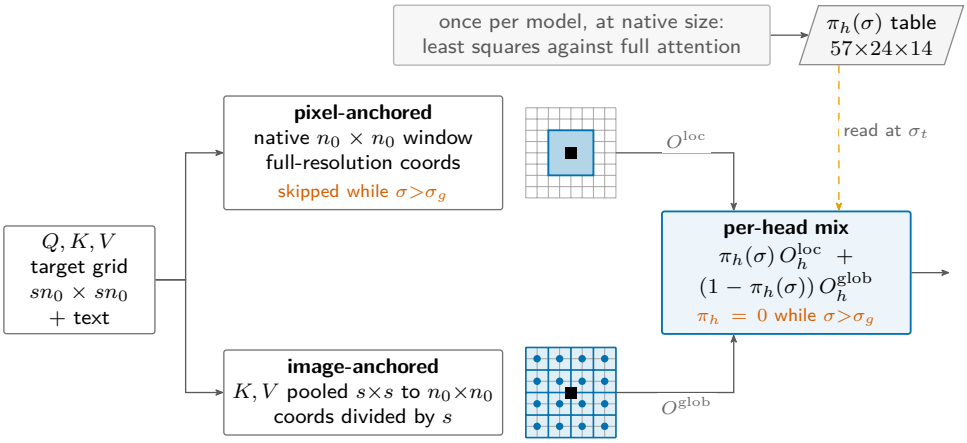


Figure 1: **SCA mixes two in-distribution receptive fields per head.** For a target grid of $sn_0 \times sn_0$ tokens, the pixel-anchored branch lets each query tile attend to a native $n_0 \times n_0$ key window with the original RoPE coordinates. The image-anchored branch subsamples keys and values by s onto an $n_0 \times n_0$ grid and divides query coordinates by s. Both see the native token count and range of relative positions. Head h mixes them with weight $\pi_h(\sigma)$ from a table calibrated once per model at native resolution. While $\sigma > \sigma_g$, $\pi_h(\sigma) = 0$ and the pixel-anchored branch is skipped. Sketches: black is a query, blue the keys it sees.

VideoGen classifies heads online (Xi et al., 2025), Sliding Tile Attention computes tiled windows efficiently (Zhang et al., 2025b), and Sparse-vDiT searches per-head sparsity offline (Chen et al., 2025). These target speed at trained resolutions. Our local branch is blocked neighborhood attention (Hassani et al., 2022) written in FlexAttention (Dong et al., 2024). Language models assign positions or spans per head (Jin et al., 2024; Zhang et al., 2024; Fu et al., 2024; Xiao et al., 2024; 2023; Peng et al., 2023), and trained vision transformers learn head-level scale splits (Yang et al., 2021; Wang et al., 2021; Pan et al., 2022; Ren et al., 2021; Liu et al., 2021). Our measurements suggest that a full-attention diffusion transformer learns a comparable split implicitly, and SCA reads it out without training.

3 Scale-Covariant Attention

3.1 Setting

A multimodal diffusion transformer such as FLUX.1 concatenates T text tokens with $N = H \times W$ image tokens and applies joint attention in every block. FLUX.1 has 19 double-stream and 38 single-stream blocks with 24 heads each, and RoPE encodes each image token's grid position $p = (r, c)$. It is trained at about 1024^2 pixels, a native grid of $n_0 \times n_0$ tokens with $n_0 = 64$. For one head, full attention is

$$O_i^{\text{full}} = \sum_j \frac{\exp(\ell_{ij})}{\sum_{j'} \exp(\ell_{ij'})} v_j, \qquad \ell_{ij} = \tfrac{1}{\sqrt{d}} \left(R_{p_i} q_i\right)^\top \left(R_{p_j} k_j\right), \tag{1}$$

with R_p the rotation for position p. The target grid is $H = s_H n_0$, $W = s_W n_0$ with integer scale factors, written s when equal. The flow-matching latent at noise level $\sigma \in [0, 1]$ is $x_\sigma = (1 - \sigma)\, x_0 + \sigma\, \epsilon$. A softmax over any partition of the keys into sets A and B splits exactly into

$$O_i^{\text{full}} = m_i\, O_i^A + (1 - m_i)\, O_i^B, \tag{2}$$

where O_i^A, O_i^B are the outputs restricted to A and B and m_i is the softmax mass on A. With A a neighborhood of the query, every head is a query-dependent mix of a near field and a far field. SCA replaces both restricted outputs with branches that stay in distribution, and the mass m_i with a calibrated per-head weight.

3.2 TWO BRANCHES, TWO COORDINATE SYSTEMS

Pixel-anchored branch. The grid is cut into query tiles of $t \times t$ tokens ($t = 8$). Every query in a tile attends to all text keys and to the $n_0 \times n_0$ window of image keys centered on the tile and clamped to the grid, with the original full-resolution positions. Each query sees $n_0^2 + T$ keys, the native count, at relative offsets below n_0 per axis, the native range. This branch models content whose extent is fixed in tokens: edges, texture, fine structure. Its output is O_i^{loc}.

Image-anchored branch. Keys and values are subsampled at the center of every $s_H \times s_W$ block, giving n_0^2 pooled image keys. Pooled key b takes the block index as its position and query i takes $\lfloor p_i/s \rfloor$ per axis, so both share one native coordinate system. Text keys are not pooled. Each query again sees $n_0^2 + T$ keys at native offsets. This branch models content whose extent is a fixed fraction of the image: layout, object count, identity. Its output is O_i^{glob}. It is AReA's first phase with the pooling factor fixed at s (Rajabi et al., 2026a).

Per-head mixing with a noise gate. For image queries of head h at noise level σ,

$$O_i \;=\; \pi_h(\sigma)\, O_i^{\mathrm{loc}} + \big(1 - \pi_h(\sigma)\big)\, O_i^{\mathrm{glob}}, \qquad \pi_h(\sigma) = 0 \text{ for } \sigma > \sigma_g. \tag{3}$$

Text queries use the image-anchored branch alone. Relative to Equation 2, $\pi_h(\sigma)$ stands in for the near-field mass; it is constant across queries, which makes it cheap and lets a few prompts determine it.

Why the gate. At noise level σ, the signal-to-noise ratio of a spatial frequency band is $((1-\sigma)/\sigma)^2 S(f)$ for image spectrum S. Natural spectra fall with frequency, so in the first steps only the lowest band carries signal, and that band fixes layout and object count. A window of w tokens completes less than one period of any frequency below $1/w$ cycles per token, so it cannot tell one large object from a repeating pattern, and neighboring windows settle the question independently. Running the mix from the first step therefore duplicates subjects at the window's scale: with a 16-token window the duplicates shrink and multiply (Figure 2). The image-anchored branch sees the whole image on the native grid and resolves this band. SCA sets $\pi_h(\sigma) = 0$ and skips the pixel-anchored branch while $\sigma > \sigma_g = 0.89$, which covers the first 8 of 28 steps at the maximum shift. This equals AReA's switch point $\sigma^\star$. Gating 13 steps ($\sigma_g = 0.80$) produces a 4×4 block mosaic, because all queries in an $s \times s$ block share one coarse position in the image-anchored branch and receive nearly the same output at high noise. We have not measured the per-band signal-to-noise ratio of FLUX.1 latents; this argument is qualitative, and the gate value comes from generations, not from a spectral calculation.

3.3 CALIBRATION BY OUTPUT MATCHING

The weights must come from where the model is trustworthy, which the target resolution is not. We emulate extrapolation inside the training distribution: at native resolution, an emulated scale s_c uses a local window of $(n_0/s_c)^2$ tokens, a pooling factor s_c onto an $(n_0/s_c)^2$ grid, and query tiles of t/s_c. The ratio of image to window is then s_c, as at a real target of scale s_c, while full attention at native resolution stays in distribution and serves as the reference. For every block, head, and noise-level bin we accumulate, over prompts, steps, image queries, and channels,

$$A = \sum \|O^{\mathrm{loc}} - O^{\mathrm{glob}}\|^2, \quad B = \sum \langle O^{\mathrm{full}} - O^{\mathrm{glob}},\, O^{\mathrm{loc}} - O^{\mathrm{glob}} \rangle, \quad E = \sum \|O^{\mathrm{full}} - O^{\mathrm{glob}}\|^2. \tag{4}$$

The squared error of the mix is $E - 2\pi B + \pi^2 A$, so the optimal weight is $\pi_h(\sigma) = \mathrm{clip}(B/A, 0, 1)$. The same statistics score any other rule (per-layer weights, one shared weight per noise level, a hard switch at any threshold, local only, global only) without another forward pass. We use 14 uniform bins in σ and interpolate linearly between bin centers; the table has $57 \times 24 \times 14$ entries. Calibration runs the unmodified model at 1024^2, collects emulated scales 2 and 4 in one pass, and took about 1.5 GPU-hours on one GH200 in total for 32 prompts at 1024^2 and 16 at 1280^2 (Appendix A).

3.4 INFERENCE AND COST

At a target with factors (s_H, s_W), SCA uses the table of the emulated scale closest in log space to $\max(s_H, s_W)$; all our targets use the scale-4 table. Algorithm 1 in Appendix B gives one layer.

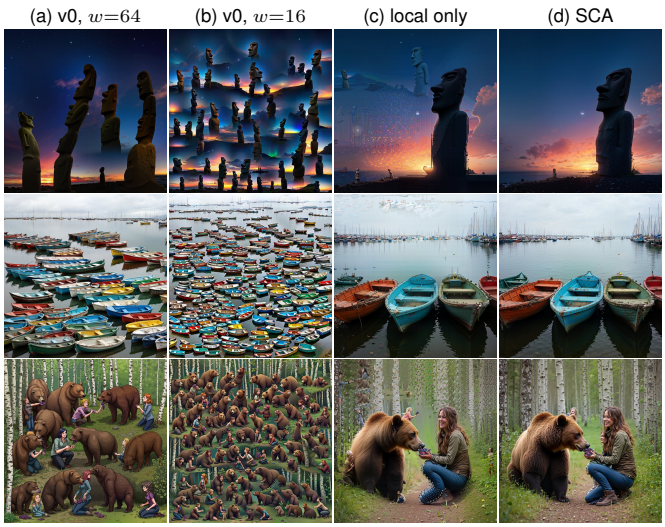


Figure 2: **Local windows at high noise duplicate subjects.** 4096^2, three development prompts, one seed per row. (a) v0, the calibrated mix from the first step with no gate: each 64-token window settles the layout on its own. (b) A 16-token window gives smaller, more numerous copies. (c) The gate (image-anchored branch alone for 8 of 28 steps) restores one main subject, but local-only attention afterwards ($\pi = 1$) leaves ghost copies and grid speckle. (d) SCA: the gate, then the per-head mix, whose global share keeps context. Small extra figures can remain (Section 6).

Weights, scheduler, VAE, and text encoders are untouched; the method is an attention-processor swap. Per head and block, full attention costs $(N+T)^2$ query-key pairs; SCA costs $N(n_0^2+T)$ for the pixel-anchored branch, skipped in 8 of 28 steps, and $(N+T)(n_0^2+T)$ for the image-anchored branch, both linear in N. At 4096^2 this is 8.4 times fewer attention FLOPs than full attention (Appendix B). The pixel-anchored branch is one FlexAttention call (Dong et al., 2024) whose block mask is the window membership itself, so it is an exact sliding window rather than an approximation.

4 What the heads want

We run the calibration on FLUX.1-dev with 32 prompts at 1024^2, guidance 4.5, 28 steps, and the shift μ at its maximum (the setting of Section 5), and repeat it with 16 prompts at 1280^2. The error measure is the relative output error $\sum \|O^{\text{full}} - O\|^2 / \sum \|O^{\text{full}}\|^2$ over image queries, pooled over blocks, heads, and steps.

Heads differ. At emulated scale 2 the standard deviation of $\pi_h(\sigma)$ over the 1368 heads is 0.24 to 0.26 at every noise level (Figure 3). Half of all (head, noise level) cells (50.4%; 49.7% at scale 4) lie in $[0.2, 0.8]$, 11.7% are above 0.95, and 2.9% below 0.05. The mean weight is 0.64 to 0.74 depending on the noise level, so the typical head leans local but still draws on the global branch. The spread is mostly within blocks: a per-layer table recovers only part of the gain of the per-head table (Figure 6 in Appendix A). The median head changes its weight by 0.18 across noise levels, and a quarter of heads by more than 0.3.

The table is a property of the model, not of the prompts. Tables fitted on two disjoint halves of the prompts correlate at 0.989 over all cells (0.986 at scale 4). Each half's table scored on the other half gives a relative error of 0.0576, against 0.0574 in sample. At 1280^2, 16 prompts give a split-half correlation of 0.974.

The weights barely depend on scale or calibration resolution. The table fitted at emulated scale 2 and applied at scale 4 gives a relative error of 0.205 against 0.200 for the scale-4 table: 2.4% higher, and still below every shared rule fitted at scale 4 (Figure 6). The scale-4 table at scale 2 gives 0.0599 against 0.0574. Across calibration resolutions, the 1024^2 table scored at 1280^2 gives 0.0589 against 0.0585, and the reverse gives 0.0578 against 0.0574. The two resolutions emulate scale 2 with windows of 32 and 40 tokens, so the weights depend on the ratio of image to window rather than on the window's absolute size. This is what we mean by scale covariance: as the image grows, each branch follows its own law and the weight between them stays put.

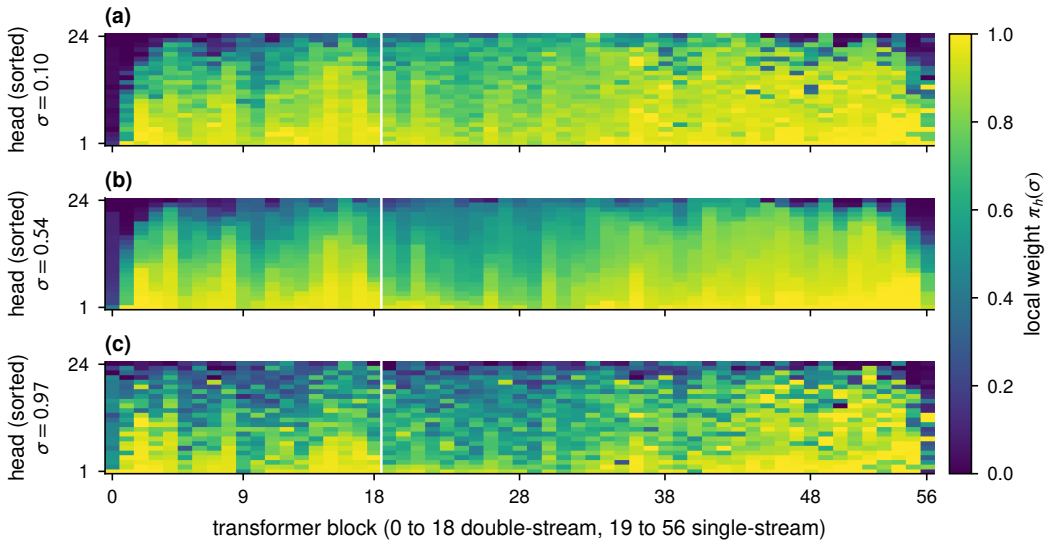


Figure 3: **Within every block, heads range from almost purely global to almost purely local.** Fitted $\pi_h(\sigma)$ for all 57 blocks and 24 heads of FLUX.1-dev at three noise levels, emulated scale 2, 32 prompts. Within each block, heads are sorted by their weight at $\sigma = 0.54$, in the same order in all panels. The white line separates double-stream from single-stream blocks.

Output error and what it misses. Per-head mixing lowers the relative error from 0.092 to 0.057 at scale 2 and from 0.261 to 0.200 at scale 4, relative to the best shared hard switch; a per-layer table reaches 0.071 and 0.223, one shared weight per noise level 0.077 and 0.230. In output energy there is no global phase: the best hard switch never switches (Appendix A). The gate of Section 3 contradicts this on purpose, because output error is dominated by texture and does not weigh the low band that sets layout at high noise.

5 Experiments

5.1 Protocol

Model, sampler, baselines. FLUX.1-dev (Black Forest Labs, 2024) in bf16, 28 flow-matching Euler steps, guidance 4.5, and the timestep shift μ at its maximum (1.15) at every resolution, with one fixed seed per prompt shared across methods and tiled VAE decoding for all. These are AReA's FLUX.1 defaults, applied to every method so that only attention differs. Baselines are full attention at the target resolution (Base) and AReA (Rajabi et al., 2026a) from its released code under this protocol, with the FLUX.1-dev checkpoint in place of the FLUX.1-Krea-dev default of the release. Appendix E gives AReA's settings and why its paper's numbers are not comparable with ours. We did not run DyPE, SEGA, UltraImage, or HRDiT under this protocol, and we ran full attention only at 4096^2.

Benchmark. The 195 evaluation prompts of Aesthetic-4K with their 4K reference images (Zhang et al., 2025a), at 4096×4096, 2048×4096, and 2048×6144 (height $\times$ width)[1]. Four of the 195 prompts (boats, Moai, woman and bear, underwater surfer) were used to choose the gate and window, so the main table reports the other 191. Ablations use a fixed subset of 46 of these.

SCA settings. $n_0 = 64$, tile $t = 8$, local window 64×64, block-center pooling, gate $\sigma_g = 0.89$, and the scale-4 table from the 32-prompt calibration at 1024^2. The calibration prompts are written

[1] AReA's paper writes its sizes without stating the order (for example 6144×2048). Its released schedules are identical under transposition, so the orientation does not change AReA's computation; our wide sizes are landscape.

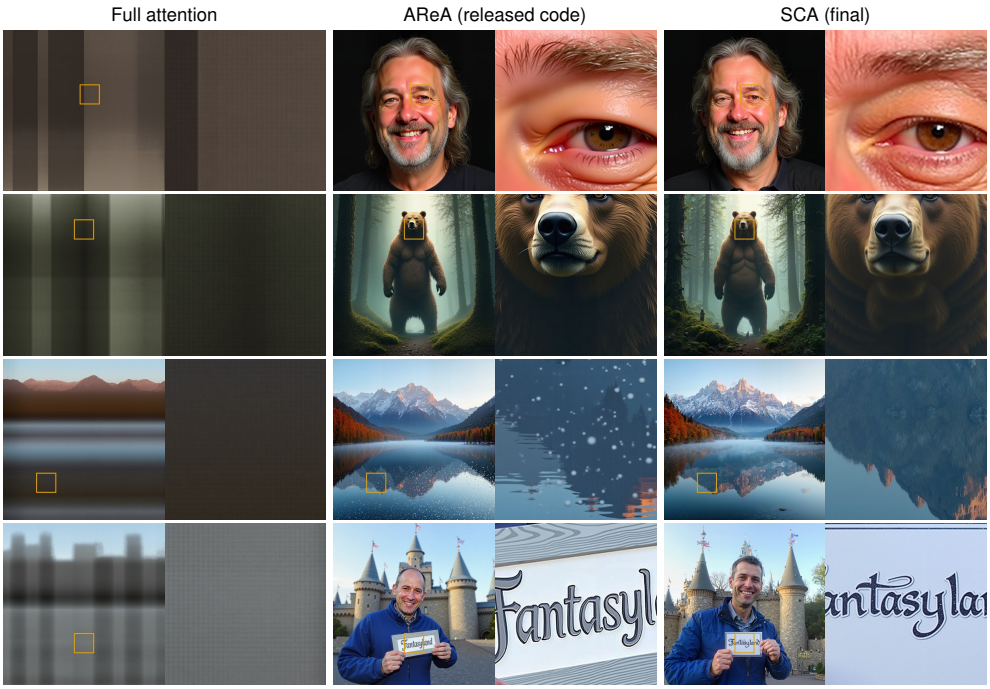


Figure 4: **At 4096^2 full attention collapses, and SCA renders finer detail than AReA.** Four Aesthetic-4K prompts outside the development set (a portrait of a gray-haired man, a standing brown bear in a forest, a mountain lake, a man holding a "Fantasyland" sign before a castle), one seed per prompt shared across methods. Each pair shows the image downscaled and the outlined 512×512 region at native resolution (1:1 pixels). Full attention gives stripes on every prompt we inspected; AReA's skin and fur are smoother, and its lake carries bright speckles.

for this purpose and share no prompt with Aesthetic-4K (the largest word-set Jaccard overlap with any evaluation prompt is 0.26). "SCA + band guard" additionally replaces the low band of each head's output by that of the image-anchored branch (Appendix D); it is the configuration of our first full run and not part of the final method.

Metrics. ImageReward (IR) (Xu et al., 2023), LAION aesthetic score (Schuhmann, 2022), CLIP score (ViT-L/14) (Radford et al., 2021), PickScore (Kirstain et al., 2023), HPSv2 and HPSv2.1 (Wu et al., 2023); FID (Heusel et al., 2017) against the 4K references at 299^2 and patch FID (FID_p) on four fixed 1024^2 crops per image; MUSIQ (Ke et al., 2021) on the resized image and on a 1024^2 crop (MUSIQ_c), CLIP-IQA (Wang et al., 2022), and ARNIQA (Agnolucci et al., 2024). Intervals are 95% percentile paired bootstraps over prompts with 1000 draws. Latency is wall-clock per image on one idle GH200 (96 GB), batch 1, including text encoding and VAE decoding, after a warm-up generation.

5.2 Main results

On 191 prompts at 4096^2, SCA beats AReA on every preference, alignment, and no-reference quality metric (Table 1). The paired differences include ImageReward $+0.31$ $[0.24, 0.38]$, HPSv2.1 $+2.50$ $[2.17, 2.88]$, PickScore $+0.73$ $[0.62, 0.86]$, CLIP score $+1.45$ $[1.06, 1.82]$, and MUSIQ_c $+10.9$ $[9.5, 12.4]$. FID falls by 7.6 $[3.5, 12.5]$ and patch KID falls significantly; patch FID falls by 3.7 $[-1.1, 7.2]$, within noise, as do KID and Inception score (6.52 vs. 6.42). At 71.0 s per image on a single GH200 (median of three images), SCA is 2.6 times faster than full attention (184.6 s) and 3.0 times faster than AReA's released code (212.6 s). The band-guard configuration also beats AReA on every metric except FID, but stays below final SCA on all but aesthetic score and patch FID and takes 17 s longer per image. Full attention collapses under this protocol: its images are fields of horizontal and vertical stripes with at most a faint trace of the subject (Figure 4), and it

Table 1: **Aesthetic-4K with FLUX.1-dev**, 191 prompts (development prompts excluded), guidance 4.5, 28 steps, one seed per prompt shared across methods. HPS, CLIP-IQA, and ARNIQA are $\times 100$. Higher is better except latency and FID. Bold: best per column within each resolution. * / †: significantly better / worse than AReA (95% paired bootstrap interval of the difference excludes zero). At 4096^2, latency is a clean measurement on a single GH200: seconds per image at batch 1, median of three images (Appendix B). ‡Mean campaign wall-clock per image on shared nodes, not a clean measurement. Full attention was run at 4096^2 only. SCA + band guard is the configuration of our first full run (Appendix D).

Method	Lat. (s)↓	IR	Aesth.	CLIP	PickS.	HPSv2	HPSv2.1	FID↓	FID_p ↓	MUSIQ	MUSIQ_c	CLIP-IQA	ARNIQA
4096×4096, scale factors (4, 4)													
Full attention	184.6	$-2.18^\dagger$	$4.51^\dagger$	$15.38^\dagger$	$17.94^\dagger$	$20.75^\dagger$	$7.61^\dagger$	$312.7^\dagger$	$257.9^\dagger$	$22.9^\dagger$	$33.2^\dagger$	$49.0^\dagger$	$54.5^\dagger$
AReA (released code)	212.6	0.92	6.28	27.67	22.35	28.64	27.20	146.0	110.2	39.4	50.4	67.5	67.3
SCA + band guard	88.1	1.20^*	**6.46***	28.87^*	22.90^*	29.40^*	28.95^*	142.3	**104.8***	41.4^*	61.0^*	70.9^*	70.8^*
SCA (final)	**71.0**	**1.23***	6.45*	**29.11***	**23.09***	**29.77***	**29.71***	**138.4***	106.5	**45.3***	**61.2***	**72.9***	**71.2***
2048×4096, scale factors (2, 4)													
AReA (released code)	$112.4^\ddagger$	**1.13**	**6.41**	**28.51**	**22.67**	28.38	**27.66**	144.8	**119.0**	51.8	61.6	**74.8**	**73.0**
SCA (final)	**38.1**‡	**1.13**	$6.21^\dagger$	28.21	22.64	**28.55***	27.46	**143.8**	$124.1^\dagger$	**55.9***	**65.9***	$73.5^\dagger$	72.8
2048×6144, scale factors (2, 6)													
AReA (released code)	$150.5^\ddagger$	0.39	**6.15**	26.49	21.51	26.41	23.35	163.6	**112.8**	42.9	53.1	67.3	69.3
SCA (final)	**55.8**‡	**0.66***	$6.00^\dagger$	**27.01**	**21.74***	**26.54***	**23.48**	**143.8***	116.5	**48.4***	**60.6***	**69.0***	**70.2***

Table 2: **Ablations of the final method at** 4096^2, 46 Aesthetic-4K prompts. The first row gives absolute values for SCA final (gate 0.89, per-head $\pi_h(\sigma)$, no band guard); other rows change one thing and report the paired difference. **Bold blue** / *orange italic*: significantly better / worse (95% paired bootstrap CI excludes zero); patch FID has no interval. HPS, CLIP-IQA, and ARNIQA are $\times 100$. "Mix err." is the relative output error of the weight rule at emulated scale 4 (Section 4). Table 4 gives the earlier ablations around the band guard.

Variant	Mix err.	IR	Aesth.	HPSv2	HPSv2.1	PickS.	CLIP	CLIP-IQA	ARNIQA	MUSIQ	MUSIQ_c	FID_p ↓
SCA final (reference)	0.200	1.12	6.45	29.71	29.98	22.96	29.41	71.3	70.5	46.1	62.5	170.2
w/o per-head mixing ($\pi = 1$)	0.261	*−0.36*	*−0.21*	*−1.78*	*−2.86*	*−1.20*	*−1.93*	*−4.6*	−1.0	*−3.5*	−0.3	+24.1
one shared $\pi(\sigma)$	0.230	*−0.09*	*−0.09*	*−0.96*	*−1.83*	*−0.38*	−0.38	*−7.7*	*−6.5*	*−4.9*	*−10.4*	−1.1
per-layer $\pi_\ell(\sigma)$	0.223	−0.05	+0.01	*−0.50*	*−0.70*	*−0.16*	−0.13	−0.9	*−1.5*	+0.7	*−2.4*	0.0
global only ($\pi = 0$)	0.627	*−0.40*	*−0.76*	*−3.32*	*−5.66*	*−1.44*	*−3.48*	*−27.0*	*−45.1*	*−3.5*	*−11.7*	+68.7
mean pooling	n/a	*−1.52*	*−0.61*	*−4.04*	*−9.75*	*−2.60*	*−3.59*	*−10.1*	*−4.8*	−1.8	*−4.2*	+17.3
v0: no gate	0.200	*−1.33*	−0.08	*−1.89*	*−4.07*	*−2.18*	*−1.65*	**+2.9**	**+4.3**	*−5.6*	+2.5	+34.7

scores ImageReward -2.18 $[-2.22, -2.12]$ and FID 312.7, significantly worse than AReA on every metric. Figure 4 also shows the difference in detail between SCA and AReA: at 1:1 scale AReA's skin and fur are smoother, and its lake surface carries bright speckles.

Non-square sizes. The advantage of SCA over AReA grows with the scale factor. It is smallest at 2048×4096, the smallest factors (2 and 4 per axis), where SCA matches AReA: ImageReward (-0.00 $[-0.08, 0.07]$), HPSv2.1, PickScore, CLIP score, ARNIQA, and FID are within noise, HPSv2 ($+0.17$), MUSIQ ($+4.1$ $[2.7, 5.7]$), and MUSIQ_c ($+4.3$) are significantly better, and aesthetic score (-0.21 $[-0.26, -0.15]$), CLIP-IQA (-1.3), and patch FID ($+5.1$ $[1.2, 9.1]$) are significantly worse. It does so at 2.9 times lower campaign time per image (38.1 s against 112.4 s). At 2048×6144, factors 2 and 6, SCA wins clearly: ImageReward $+0.27$ $[0.15, 0.40]$, FID -19.8 $[-24.5, -11.9]$, KID -0.0096, Inception score $+0.99$, PickScore $+0.23$, MUSIQ $+5.5$, MUSIQ_c $+7.5$, CLIP-IQA $+1.6$, ARNIQA $+0.9$, and HPSv2 $+0.13$ are all significant; HPSv2.1, CLIP score, and patch FID are within noise; aesthetic score is significantly worse (-0.15 $[-0.25, -0.04]$). Campaign time is 2.7 times lower (55.8 s against 150.5 s). From 2048×4096 to 4096^2 to 2048×6144, the FID difference goes from -1.1 (within noise) to -7.6 to -19.8, and AReA's own ImageReward falls to 0.39 at the largest factor while SCA keeps 0.66. Both non-square sizes use the scale-4 table; the scale-2 table changes the 2048×4096 results negligibly (Appendix C).

5.3 Ablations

Table 2 changes one component of the final method at a time; all rows except v0 keep the gate. Table 4 in Appendix D holds the ablations we ran earlier around the band-guard configuration.

The per-head mix supplies global context after the gate. Setting $\pi = 1$ after the gate, so that every head attends only inside its native window for the last 20 steps, lowers ImageReward by 0.36 $[0.21, 0.52]$, HPSv2.1 by 2.86 $[2.36, 3.35]$, PickScore by 1.20 $[0.99, 1.44]$, CLIP score by 1.93 $[1.20, 2.68]$, aesthetic score by 0.21, CLIP-IQA by 4.6, and MUSIQ by 3.5, all significant, and raises patch FID from 170.2 to 194.3; only ARNIQA and MUSIQ_c stay within noise. Most heads lean local (mean weight 0.64 to 0.74), but their calibrated share of the image-anchored branch keeps the image coherent after the opening phase. With the band guard, local only matched the mix, because the guard injects the image-anchored branch's low band into every head whatever its weight (Appendix D); without it the mix is the only route for global information after the gate.

The opening phase is essential. Without the gate (v0, the calibrated mix from the first step), ImageReward falls by 1.33 $[1.06, 1.60]$ against final SCA and patch FID rises by 34.7: subjects repeat across the image (Figure 2). Yet v0 scores higher on CLIP-IQA ($+2.9$) and ARNIQA ($+4.3$), both significant, because each repeated subject is sharp; no-reference quality metrics do not see layout.

The global branch cannot work alone. Global only loses on every metric (patch FID $+68.7$, ARNIQA -45): with keys s tokens apart it cannot render detail. Mean pooling blurs the image-anchored keys (ImageReward -1.52), confirming AReA's choice of block-center subsampling.

Per-head weights beat shared and per-layer weights. Without the guard, replacing the per-head table with one shared weight per noise level lowers ImageReward by 0.09 $[0.01, 0.17]$, HPSv2.1 by 1.83, PickScore by 0.38, CLIP-IQA by 7.7, ARNIQA by 6.5, and MUSIQ_c by 10.4, all significant. Per-layer weights recover part of this gap, as the output error predicts (0.223 against 0.230 and 0.200), but still lose significantly on HPSv2.1 (-0.70), PickScore (-0.16), ARNIQA (-1.5), and MUSIQ_c (-2.4); their ImageReward drop (-0.05) is within noise.

6 Limitations

Small scale factors. At 2048×4096 SCA only matches AReA and is worse on aesthetic score, CLIP-IQA, and patch FID; the speed gain remains. **Residual duplication.** With the gate alone, small extra figures can appear late in sampling on some prompts, and the underwater surfer prompt keeps doubled limbs in every variant we tried; AReA's image for that prompt shows a large translucent ghost. The band guard (Appendix D) removes some of these at a cost. **The gate threshold** equals AReA's $\sigma^\star$, was compared only with 0.80 on four prompts, and its dependence on the shift and step count is untested. **Calibration objective.** Squared output error misses low-band errors, so the gate is set by hand. **Emulation.** Emulated scale 4 at 1024^2 uses a 16-token window, the real 4096^2 target a 64-token window. Transfer across emulated scales supports, but cannot prove, that the weights depend only on the ratio. **Scope.** All results are for FLUX.1-dev. We have not calibrated or evaluated a second multimodal diffusion transformer, so the claim that receptive-field needs are head-specific and scale-stable is shown for one model.

7 Conclusion

At its training resolution, FLUX.1-dev shows how each attention head splits between a neighborhood fixed in tokens and a view of the whole image; the split differs between heads, holds across prompts, and changes little with scale, so it can be measured once and reused at 4K. Output matching sets the weights but not the opening phase, in which every head must see the whole image. With both, SCA beats AReA's released code at 4096^2 (ImageReward 1.23 against 0.92, FID 138.4 against 146.0) while running 3.0 times faster, ties it at 2048×4096, and beats it at 2048×6144 with the largest FID gain (-19.8).

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A Calibration details

Statistics. For every emulated scale, block, head, and noise-level bin, the pass stores A, B, E, the error of the local branch alone, $\sum \|O^{\text{full}}\|^2$, and the query count, in float64, summed over image queries and channels. Statistics are written per prompt and merged by summation, so prompts can be added later and split-half fits need no extra pass. Text-query outputs are not in the objective, because SCA sends text queries through the image-anchored branch only.

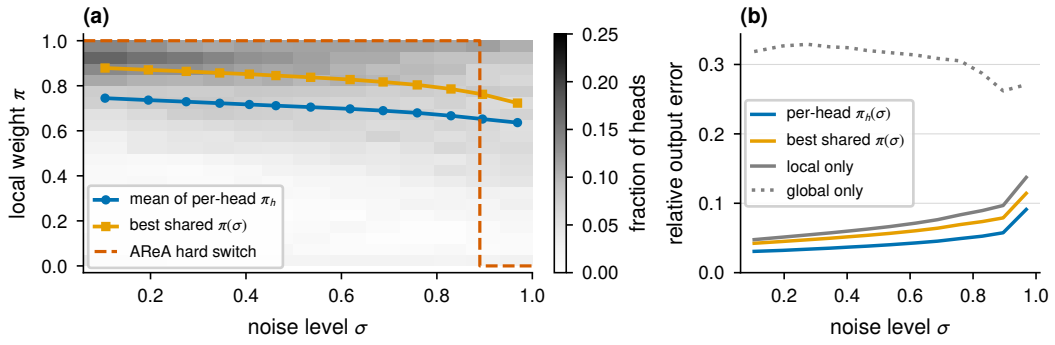


Figure 5: **In output energy, no noise level prefers the global branch.** (a) Distribution of $\pi_h(\sigma)$ over the 1368 heads per noise-level bin (grey), the mean over heads, and the best single shared weight, at emulated scale 2. The dashed step is AReA's schedule read as a weight. (b) Relative output error per bin. The local branch alone beats the global branch alone in every bin, and per-head mixing beats both. 32 prompts, one calibration run.

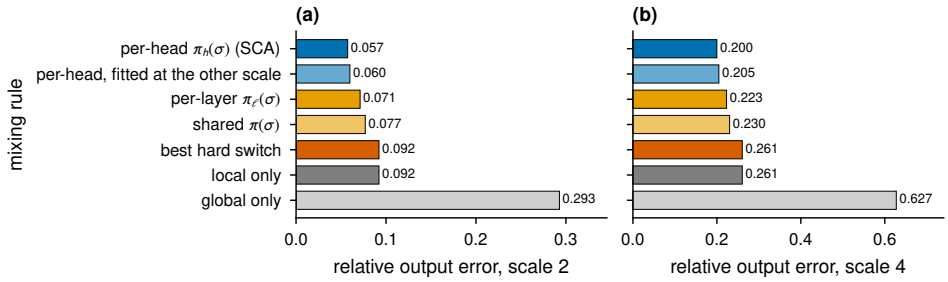


Figure 6: **Per-head weights explain full attention best, and transfer across scales.** Relative output error against full attention for each mixing rule at emulated scales 2 (a) and 4 (b), 32 prompts. "Fitted at the other scale" applies the scale-4 table at scale 2 and vice versa. The best hard switch puts its threshold at $\sigma^\star = 1$ and so equals local only. In-sample and held-out errors agree within 0.0003.

Geometry at emulated scale. At 1024^2 the grid is 64×64. Emulated scale 2 uses a 32×32 window, 2×2 pooling onto a 32×32 grid, and 4×4 query tiles; emulated scale 4 uses a 16×16 window, 4×4 pooling onto a 16×16 grid, and 2×2 tiles. At 1280^2 the grid is 80×80, and emulated scale 2 uses a 40×40 window and 2×2 pooling onto a 40×40 grid.

Noise-level bins. Bins have width $1/14$. With 28 steps and $\mu = 1.15$, σ runs from 1.0 to 0.105, so the lowest bin is empty and copies its neighbor; 13 bins are populated. Bin centers are the mean σ of their steps. Under this schedule 8 steps have $\sigma > 0.89$ (the ninth has $\sigma = 0.888$) and 13 have $\sigma > 0.80$.

No global phase in output energy. The best single shared weight falls from 0.88 at $\sigma = 0.10$ to 0.72 at $\sigma = 0.97$, and the mean per-head weight from 0.74 to 0.64 (Figure 5). The global share rises with noise, but at no noise level does the average head prefer the global branch. Among hard switches from global to local, the best threshold is $\sigma^\star = 1$, which equals local only. A switch at $\sigma^\star = 0.89$ between our two branches raises the error from 0.092 to 0.152 at scale 2 and from 0.261 to 0.376 at scale 4. In the noisiest bin, the global branch alone has relative error 0.27 at scale 2, the local branch alone 0.14, and the per-head mix 0.09. The generation results of Section 5.3 show the opposite preference for the opening steps, which is the gap between squared output error and layout.

Attention mass. For 256 random image queries per block and step, the pass also records the softmax mass on text keys and on image keys within Chebyshev distance 2, 4, 8, and 16. At 1024^2, averaged over heads and queries, text receives 15 to 19% of the mass; image keys within distance

Algorithm 1 One SCA attention layer at noise level σ (per head h).

Require: pre-RoPE Q, K, V for T text and $N = H \times W$ image tokens; factors (s_H, s_W); table $\pi_h(\sigma)$; gate $\sigma_g = 0.89$

$\tilde{K}, \tilde{V} \leftarrow$ image keys and values at the center of every $s_H \times s_W$ block, plus all text keys

$O^{\text{glob}} \leftarrow \text{Attn}\big(Q \text{ at } \lfloor p/s \rfloor,\ \tilde{K} \text{ at block index},\ \tilde{V}\big)$ ▷ all queries, text included

if $\sigma > \sigma_g$ **then return** O^{glob} ▷ opening phase: pixel-anchored branch skipped

$O^{\text{loc}} \leftarrow$ FlexAttention of each $t \times t$ image-query tile over text keys and its $n_0 \times n_0$ window, original RoPE

return $[\, O^{\text{glob}}_{\text{text}},\ \pi_h(\sigma)\, O^{\text{loc}} + (1 - \pi_h(\sigma))\, O^{\text{glob}}_{\text{img}} \,]$ ▷ mix in fp32

16 receive 52% at $\sigma = 0.10$ and 40% at $\sigma = 0.97$, and keys outside that 33×33 neighborhood still receive 29% and 41%. A purely local receptive field cannot account for any step even at native resolution.

B Implementation details and cost

FlexAttention window. Image tokens are reordered tile-major; query blocks cover 8×16 tokens and key blocks 8×8, shifted so that unclamped windows start on a block edge, and the mask function is the window membership itself. The kernel matches a reference that gathers each tile's key window explicitly to within one bf16 unit in the last place, and both are equally far from an fp32 reference. RoPE, pooling, both branches, and the mix compile into one graph per layer.

Paired intervals for Table 2. The intervals of every row in Table 2 are paired bootstraps over the 46 prompts with 2000 draws and seed 0, computed by `scripts/paired_bootstrap.py` from the per-image scores against final SCA; the outputs are `results/ablnb_<method>_deltas.csv`. Patch FID is a set-level metric (`results/ablnb_base_summary.csv`) and has no paired interval.

Position ids in bf16. The FLUX.1 pipeline builds image position ids in bf16, which cannot represent every integer above 256, so positions beyond 256 tokens per axis are rounded. This affects every method, including full attention and AReA, at the 2048×6144 setting (384 tokens per row). SCA takes the grid size from the sampler instead of inferring it from the rounded ids, so that its window and pooling geometry match the true grid.

Table 3: **Cost.** Top: analytic attention FLOPs per image (57 blocks, 24 heads, head dimension 128, 28 steps, $T = 512$); "gated" skips the pixel-anchored branch in the first 8 steps; ratio is full over gated SCA. Bottom: measured time of one attention layer in bf16 on GH200, minimum over interleaved runs; SCA is both branches and the mix.

Resolution ($H \times W$)	Grid	(s_H, s_W)	Full	SCA, no gate	SCA, gated	Ratio
2048×4096	128×256	(2, 4)	2.17×10^{16}	5.97×10^{15}	5.12×10^{15}	4.2
4096×4096	256×256	(4, 4)	8.56×10^{16}	1.19×10^{16}	1.02×10^{16}	8.4
2048×6144	128×384	(2, 6)	4.84×10^{16}	8.93×10^{15}	7.66×10^{15}	6.3

Token grid	Pixels	Full (SDPA, ms)	SCA (ms)	Speedup
128×128	2048^2	7.1	5.2	1.4
256×256	4096^2	87.3	18.7	4.7
384×128	6144×2048	49.6	14.0	3.5

End-to-end latency. On a single GH200, batch 1, 4096^2, 28 steps, after a warm-up generation, we report the median over three images (all timings in `results/latency_4096_batch1.csv`): full attention 184.6 s (184.1, 184.7, 184.6), AReA (released code) 212.6 s (207.3, 225.8, 212.6), SCA + band guard 88.1 s (88.1, 98.7, 88.1; single-pass compiled band, outputs equal to the campaign code up to floating-point rounding), and SCA final 71.0 s (71.0, 83.1, 70.9). On the shared campaign nodes, per-image times were 71 s for SCA final, 120 s for SCA + band guard, and 262 s for AReA; the campaign code used an unfused band, so dropping it saved more there (48 s) than in the clean

Table 4: **Ablations around the band-guard configuration at** 4096^2, 46 Aesthetic-4K prompts. The reference row gives absolute values for SCA + band guard (band 16); every other row changes one thing, keeps the guard unless stated, and reports its paired difference to the reference. Marks and units as in Table 2; here patch FID also has paired intervals.

Variant	Mix err.	IR	Aesth.	HPSv2	HPSv2.1	PickS.	CLIP	CLIP-IQA	ARNIQA	MUSIQ	MUSIQ_c	$\text{FID}_p \downarrow$
SCA + band guard (reference)	0.200	1.17	6.43	29.18	28.81	22.71	29.01	69.6	69.6	43.1	62.1	169.3
no band (**= SCA final**)	0.200	−0.04	+0.01	+0.54	+1.17	+0.25	+0.40	+1.7	+0.9	+2.9	+0.4	+0.8
no gate	0.200	−0.36	−0.23	−0.49	−1.39	−0.86	−1.15	−0.6	+1.6	−0.7	−1.0	+9.0
one shared $\pi(\sigma)$	0.230	−0.03	−0.06	−0.20	−0.33	+0.02	−0.09	−6.5	−4.6	+0.5	−5.7	−4.8
per-layer $\pi_\ell(\sigma)$	0.223	−0.02	+0.03	−0.08	−0.01	+0.01	−0.03	−1.2	−1.2	−0.3	−1.5	−1.4
local only ($\pi = 1$)	0.261	−0.01	+0.03	+0.16	+0.55	+0.12	+0.16	+0.5	+0.6	+1.0	+0.9	+1.3
hard routing, threshold 0.5	n/a	−0.05	−0.02	+0.02	+0.18	+0.04	0.00	+0.3	+0.1	+1.2	+0.5	+1.5
hard routing, threshold 0.3	n/a	−0.03	+0.02	+0.11	+0.34	+0.09	+0.13	+0.6	+0.6	+0.8	+0.8	+1.1
mean pooling	n/a	−0.85	−0.39	−2.35	−5.81	−1.80	−2.96	−5.4	−1.1	−1.8	−0.8	+18.8

measurement (17 s). At the non-square sizes we have only campaign timings on shared nodes (mean per image over 191 prompts): 2048×4096, SCA 38.1 s and AReA 112.4 s; 2048×6144, SCA 55.8 s and AReA 150.5 s. These are not clean measurements, but the two methods ran on the same pool of nodes.

C Calibration table at non-square sizes

At $(s_H, s_W) = (2, 4)$ the two axes have different factors, and SCA uses the table of the emulated scale nearest the larger factor, here scale 4 (Section 3.4). On the 46 ablation prompts at 2048×4096, replacing it with the scale-2 table changes the paired metrics negligibly: HPSv2 $+0.09$ $[0.04, 0.13]$, HPSv2.1 $+0.17$ $[0.08, 0.26]$ (both $\times 100$), and PickScore $+0.05$ $[0.00, 0.09]$ are significant but small, and ImageReward ($+0.01$ $[-0.03, 0.05]$), aesthetic score, CLIP score, MUSIQ, MUSIQ_c, CLIP-IQA, and ARNIQA are within noise (`results/ps2_abl_ps2_vs_scanb.csv`). The choice of table therefore does not explain why SCA only matches AReA at this size, and we keep the default rule.

D The low-band duplication guard

Construction. Let LP_p average-pool a head's image output over $p \times p$ tokens and upsample it bilinearly, and $\text{HP}_p = \text{I} - \text{LP}_p$. The guard replaces the low band of the mix M by that of the image-anchored branch:

$$O \;=\; \text{LP}_p\big(O^{\text{glob}}\big) + \text{HP}_p\big(M\big) \;=\; O^{\text{glob}} + \pi_h(\sigma)\,\text{HP}_p\big(O^{\text{loc}} - O^{\text{glob}}\big). \tag{5}$$

The pixel-anchored branch then contributes only its difference from the image-anchored branch above the cutoff. The cutoff should lie below $1/(2s)$, where the image-anchored branch aliases, and above $1/w$, below which the window cannot resolve; with $w = 64$ and $s = 4$ this leaves periods of 8 to 64 tokens. We used $p = 16$; $p = 8$ and $p = 32$ were worse on four development prompts, and a noise-dependent cutoff behaved like $p = 16$. FreeScale (Qiu et al., 2024) also fuses two self-attention outputs by frequency band, in the opposite direction: it keeps the Gaussian-blurred low band of attention over overlapping crops and takes the high band from full self-attention at the target size, to suppress small repeated objects. The guard instead takes the low band from the image-anchored branch, which stays at native token count and offsets, and keeps the per-head weight on the high band.

What it fixes and what it costs. On the development prompts, the band alone (no gate) fixes the boats but leaves extra statues and faces on the Moai and woman-and-bear prompts; the gate alone gets the main subject right but small extra figures appear late on some prompts; gate and band together give one subject on the boats, Moai, and woman-and-bear prompts. On 46 prompts, however, removing the guard improves HPSv2 ($+0.54$), HPSv2.1 ($+1.17$), PickScore ($+0.25$), CLIP-IQA ($+1.7$), and MUSIQ ($+2.9$), all significant, leaves ImageReward, aesthetic score, ARNIQA, MUSIQ_c, and patch FID within noise, and saves 17 s per image on an idle GPU. The guard also hides what the per-head mix contributes. With it (Table 4), local only matched the per-head mix within

2% on preference metrics and was slightly higher on HPSv2.1 ($+0.55$), because the guard replaces the low band of every head's output with that of the image-anchored branch whatever the head's weight, so it injects global low frequencies even at $\pi = 1$; without the guard local only loses on nearly every metric (Section 5.3). Likewise, with the guard one shared weight and per-layer weights matched the per-head table on preference metrics and lost only on no-reference quality, and hard per-head routing (10 to 17% of heads global at threshold 0.5, about 6% at 0.3) stayed within 2% of the mix on preference metrics. The guard is therefore an option for prompts where late duplication is visible, not part of the default method.

E REPRODUCING AREA

We run AReA's released code with its FLUX.1 defaults (block-center pooling, first phase for 8 of 28 steps, halo width 8, bridge ratio 0.5), the FLUX.1-dev checkpoint, guidance 4.5, 28 steps, μ at its maximum, and the seeds shared with the other methods. AReA's paper reports ImageReward 1.32 at 4K under its own settings; our shared protocol gives 0.92, so cross-paper numbers are not comparable, while all numbers in this paper come from one protocol. AReA's paper reports 151 s for AReA and 314 s for full attention at 4096^2 on an H100; we measure 212.6 s and 184.6 s on a single GH200 (medians of three images). We use commit `9e38ec2` of the released repository (github.com/KooroshRH/AReA), which includes the cost cap on halo and bridge tokens and the jagged halo edge; neither appears in the paper text, and the paper does not say whether its Table 1 used them. Recomputed from the code with 512 text tokens, the schedules are as follows. At 4096×4096 and 2048×4096, the first-phase pooling factors are 4, 4, 4, 4, 2, 2, 2, 2; at 2048×6144 they are 8, 8, 8, 4, 4, 4, 2, 2. The second-phase windows shrink over 20 steps from 180, 151, and 192 tokens respectively to 64. At 2048×6144 no window is cheaper than full attention once the text tokens are counted, so the first window falls back to 192 tokens and the cap drops the bridge and halo on the first three second-phase steps. Other differences from the release defaults: the FLUX.1-dev checkpoint instead of FLUX.1-Krea-dev, and one seed per prompt shared across methods instead of one seed for all prompts.